

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1228

Henry D King 100 kV
May 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1228. The Interconnection Customer proposes a 117.9 MW battery energy storage facility interconnecting at the Henry D King 100 kV substation in the Illinois area. The request was received on June 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 320 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,181,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dpc Oakdale - Carrizo Plains 100 KV	Basscreek - Wallisville 100 KV	198.4	112	100.0
Madison Park - Dpc Oakdale 100 KV	Wallisville - Basscreek 100 KV	174.0	113	100.0
Basscreek - Wallisville Substation 100 KV	Carrizo Plains - Dpc Oakdale 100 KV Line	159.9	139	100.0
Henry D King - Basscreek 100 KV L	Dpc Oakdale - Carrizo Plains 100 KV	148.0	135	100.0
Henry D King - Madison Park 100 KV	Basscreek - Wallisville 100 KV	131.6	128	100.0
N Knox - Carrizo Plains 100 KV	Wallisville - Basscreek 100 KV L	112.9	137	41.9
Wallisville - Lullwater 100 KV #2	Lullwater - Wallisville Substation 100 KV Ckt 1	109.3	158	24.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dpc Oakdale - Carrizo Plains 100 KV	Rebuild line with higher-capacity conductor	\$2,472,000	100% (\$2,472,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dpc Oakdale - Madison Park 100 KV	Rebuild line with higher-capacity conductor	\$826,000	100% (\$826,000)
Basscreek Substation - Wallisville 100 KV	Rebuild line with higher-capacity conductor	\$4,258,000	100% (\$4,258,000)
Basscreek - Henry D King 100 KV	Rebuild line with higher-capacity conductor	\$1,280,000	100% (\$1,280,000)
Henry D King - Madison Park 100 KV	Rebuild line with higher-capacity conductor	\$836,000	100% (\$836,000)
N Knox Substation - Carrizo Plains 100 KV line	Rebuild line with higher-capacity conductor	\$3,802,000	74% (\$2,806,000)
Wallisville - Lullwater 100 KV #2	Reconductor line	\$1,703,000	100% (\$1,703,000)
	Total allocated to project		\$14,181,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.